EXP 6

CODE:

```
#include <Servo.h>
const int trigPin = 9;
const int echoPin = 10;
const int servoPin = 6;

Servo myServo;

const int detectionDistance = 50;

void setup() {
  Serial.begin(9600);

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  myServo.attach(servoPin);
  myServo.write(0);
  delay(1000);
}

void loop() {
  long duration;
  int distance;
  digitalWrite(trigPin, LOW);
```

```
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);

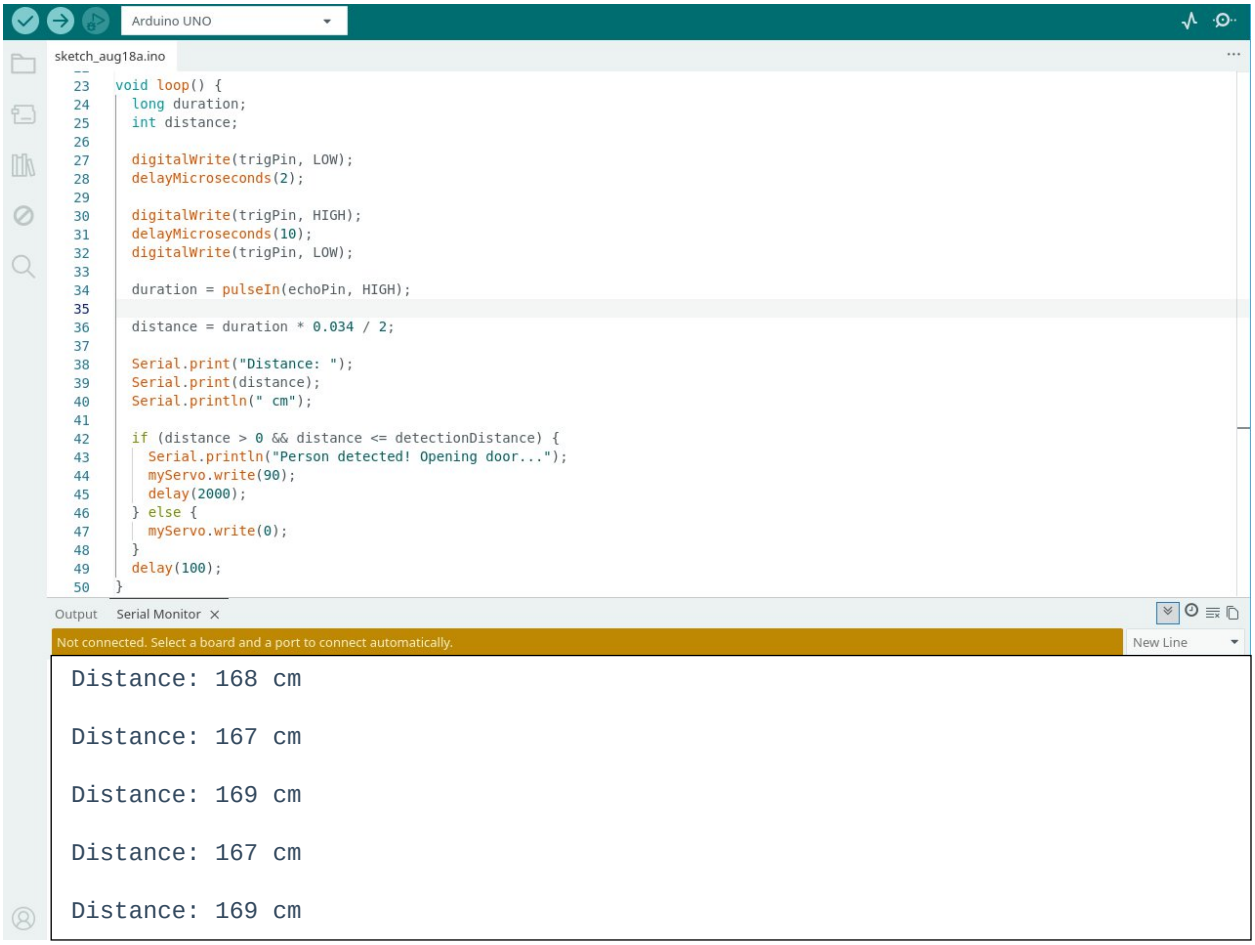
  // Calculate the distance in cm (Speed of sound is 0.034 cm/us, divided by 2 for round trip)
  distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  if (distance > 0 && distance <= detectionDistance) {
    Serial.println("Person detected! Opening door...");
    myServo.write(90);
    delay(2000);
  } else {
    myServo.write(0);
  }

delay(100);
}
```

OUTPUT:


Arduino UNO
sketch_aug18a.ino
void loop() {
long duration;
int distance;
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
duration = pulseIn(echoPin, HIGH);
distance = duration * 0.034 / 2;
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");
if (distance > 0 && distance <= detectionDistance) {
Serial.println("Person detected! Opening door...");
myServo.write(90);
delay(2000);
} else {
myServo.write(0);
}
delay(100);
}
Output
Serial Monitor
Not connected. Select a board and a port to connect automatically.
New Line
Distance: 168 cm
Distance: 167 cm
Distance: 169 cm
Distance: 167 cm
Distance: 169 cm

